

Name: _____



Converting fractions to recurring decimals

Task A: Recognising recurring decimals

1) Match the correctly notated **recurring decimal** with the decimal form.

$0.\dot{7}$	$0.75757575\dots$
$0.\dot{7}5\dot{4}$	$0.7555555\dots$
$0.\dot{7}\dot{5}$	$0.75454545\dots$
$0.7\dot{5}\dot{4}$	$0.777777\dots$
$0.7\dot{5}$	$0.75475475\dots$

2) Using short division, identify the **recurring decimal** of the following fractions:

- a) $\frac{1}{9}$
- b) $\frac{5}{9}$
- c) $\frac{1}{24}$

3) Circle the fractions which are equivalent to a **recurring decimal**.

$$\begin{array}{ccccc} \frac{3}{5} & \frac{6}{7} & \frac{12}{5} & \frac{3}{60} & \frac{11}{4} \\ \\ \frac{7}{6} & \frac{9}{12} & \frac{5}{12} & \frac{4}{11} & \end{array}$$

Task B: Using a calculator

1) Using a calculator, match the fraction to the equivalent decimal to the correct statement.

$$\frac{59}{64}$$

$$0.320754717$$

Terminating decimal

$$\frac{17}{53}$$

$$3.14159292$$

Non-terminating decimal

$$\frac{355}{113}$$

$$0.921875$$

2) Match the correctly notated **recurring decimal** with the fractional form.

$$0.\dot{4} \quad 0.\dot{7}\dot{5} \quad 0.7\dot{5} \quad 0.\dot{7} \quad 0.\dot{7}5\dot{4} \quad 0.7\dot{5}\dot{4}$$

$$\frac{83}{110}$$

$$\frac{754}{999}$$

$$\frac{25}{33}$$

$$\frac{4}{9}$$

$$\frac{34}{45}$$

$$\frac{7}{9}$$

3) Write the following **recurring decimals** as a fraction.

a) $0.124\dot{5}$

b) $3.424\dot{8}$

c) $3.1\dot{4}$